

Chapter 3

Logic Gates (逻辑门)

Objectives:

Basic gates: NOT (非), AND (与), OR (或)

Extend gates: NAND (与非), NOR (或非),

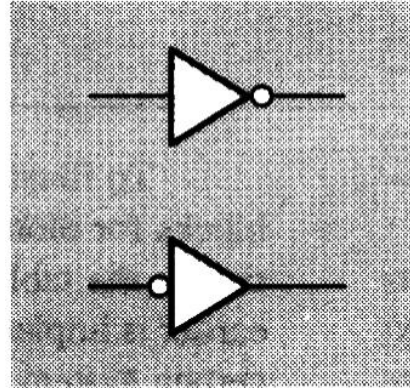
Exclusive-OR (异或), Exclusive-NOR (同或)

Symbols

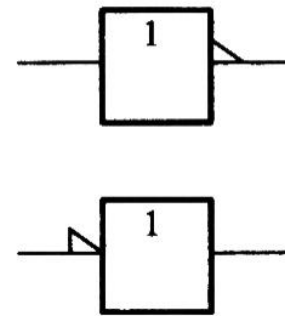
- Logic Gate (逻辑门) is the smallest operational function unit in a digital system, which will perform the basic operations.
(在数字系统中，逻辑门是最小的功能单元)

3-1 THE INVERTER (反相器)

- The **inverter** performs the operation called **inversion (反相)** or **complementation (反码)**.
- The polarity indicator is a "bubble". When appearing on the input/output, it means that a LOW level is the active or asserted input/output state.



(a) Distinctive shape symbols
with negation indicators



(b) Rectangular outline symbols with
polarity indicators

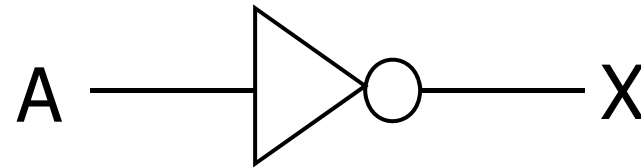
FIGURE 3-1

Standard logic symbols for the inverter (ANSI/IEEE Std. 91-1984).

The Inverter

- **Symbol**

(符号)



- **Logic expression**

(逻辑表达式)

$$X = \bar{A}$$

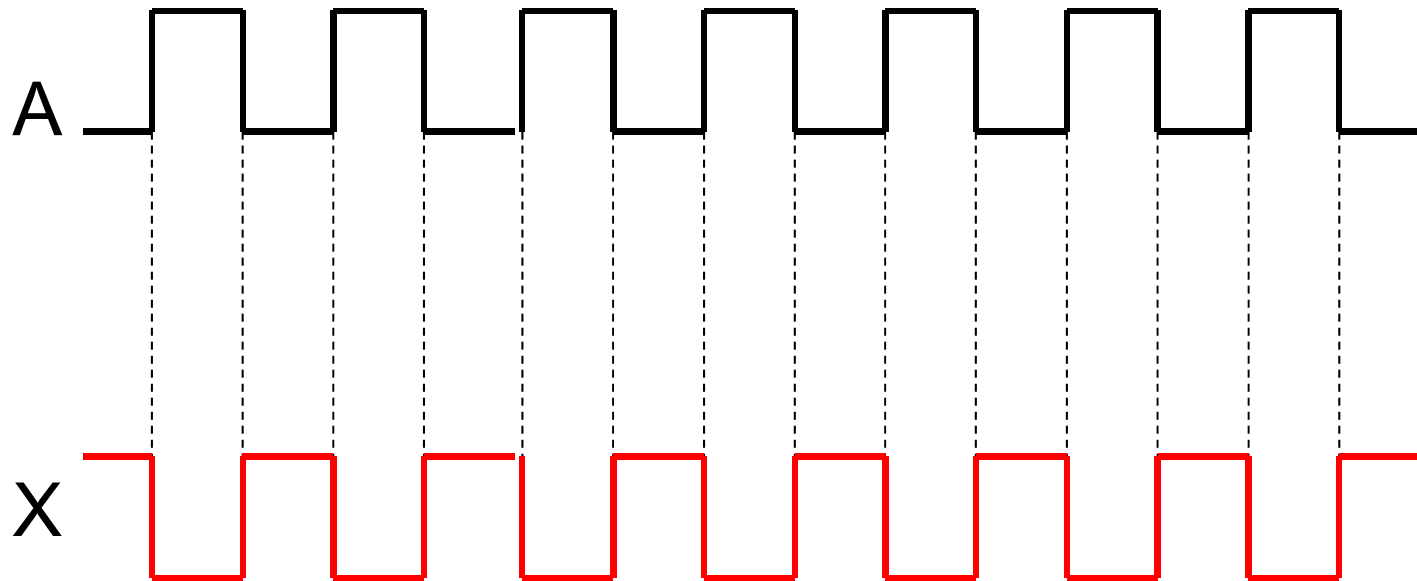
- **Truth table**

(真值表)

Input A	Output X
0	1
1	0

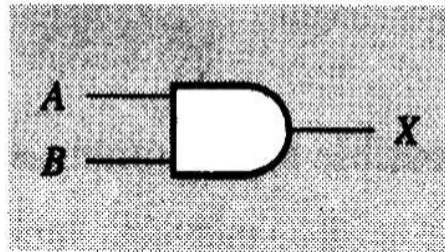
The complemented variable $\bar{A}$ can be read as “A bar” or “not A.”

• *Waveform*

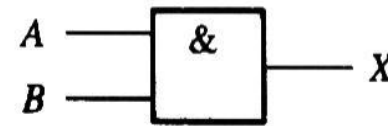


3-2 THE AND GATE (与门)

- The AND gate produces a HIGH only when all of the inputs are HIGH. When any of the inputs is LOW, the output is LOW.
- The basic purpose of an AND gate is to determine when certain conditions are simultaneously true.



(a) Distinctive shape



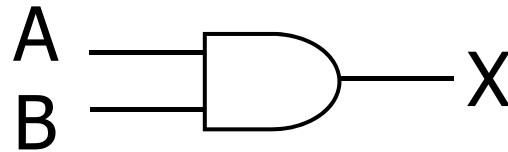
(b) Rectangular outline with the AND (&) qualifying symbol

FIGURE 3-8

Standard logic symbols for the AND gate showing two inputs (ANSI/IEEE Std. 91-1984).

The AND Gate

•*Symbol*



•*Logic expression*

$$X = AB$$

•*Truth table*

Inputs		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

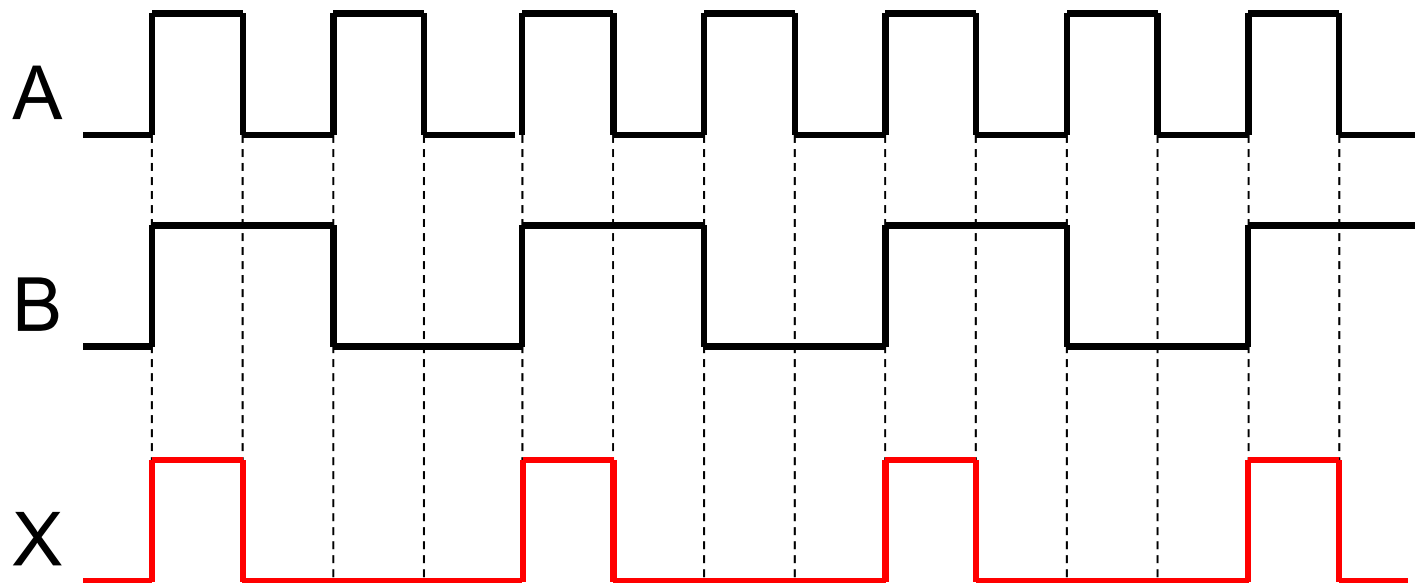
The AND Gate

- Boolean multiplication/1-binary multiplication is the same as the AND

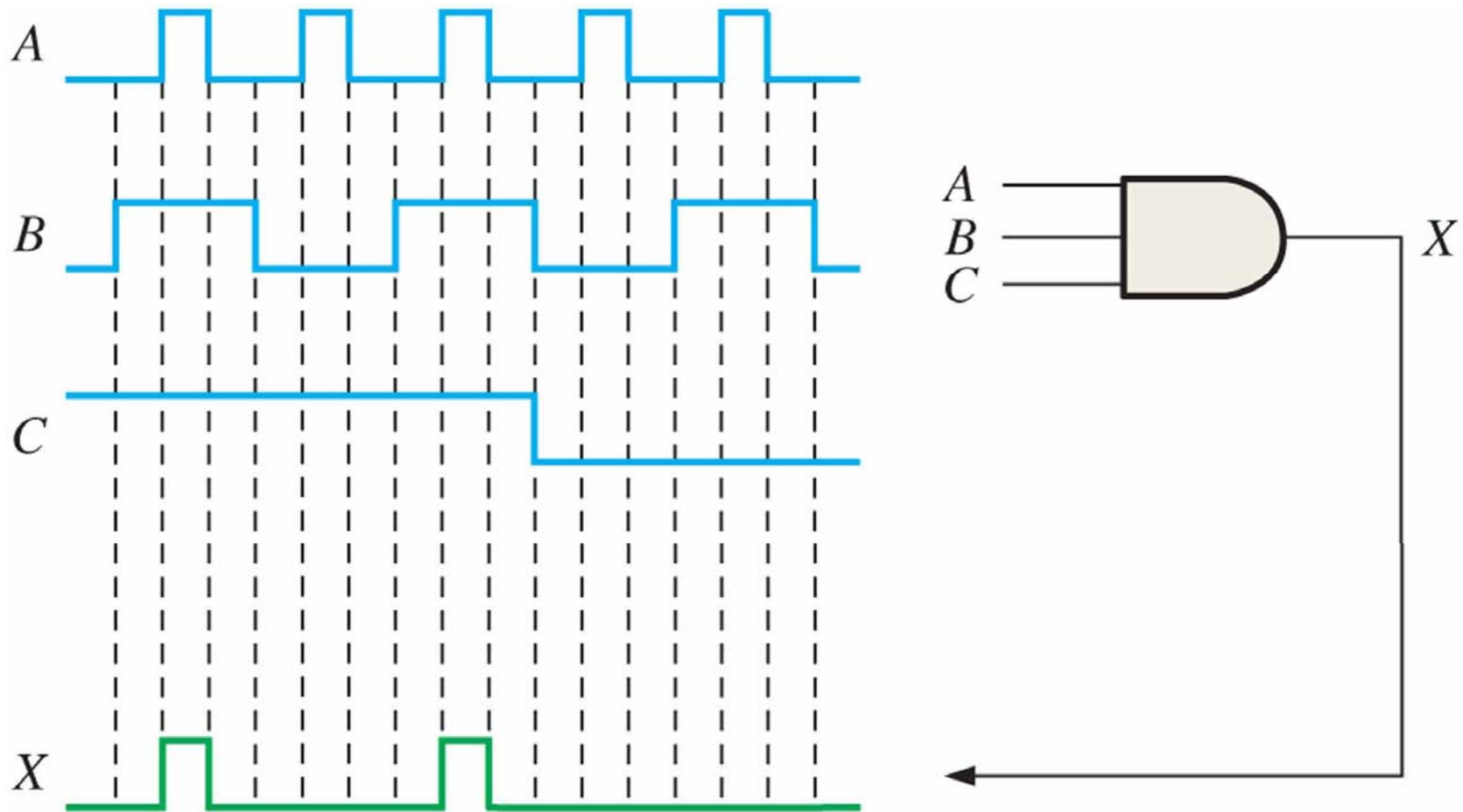
TABLE 3-4

<i>A</i>	<i>B</i>	<i>AB = X</i>
0	0	$0 \cdot 0 = 0$
0	1	$0 \cdot 1 = 0$
1	0	$1 \cdot 0 = 0$
1	1	$1 \cdot 1 = 1$

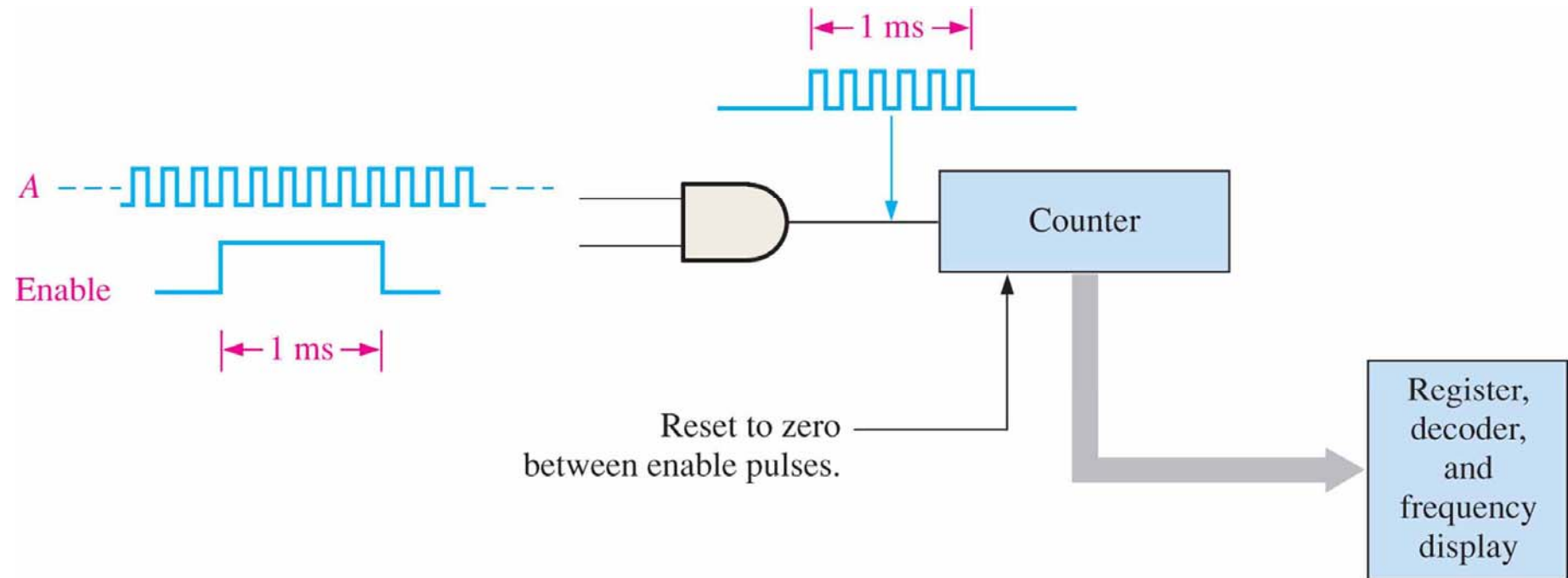
• *Waveform*



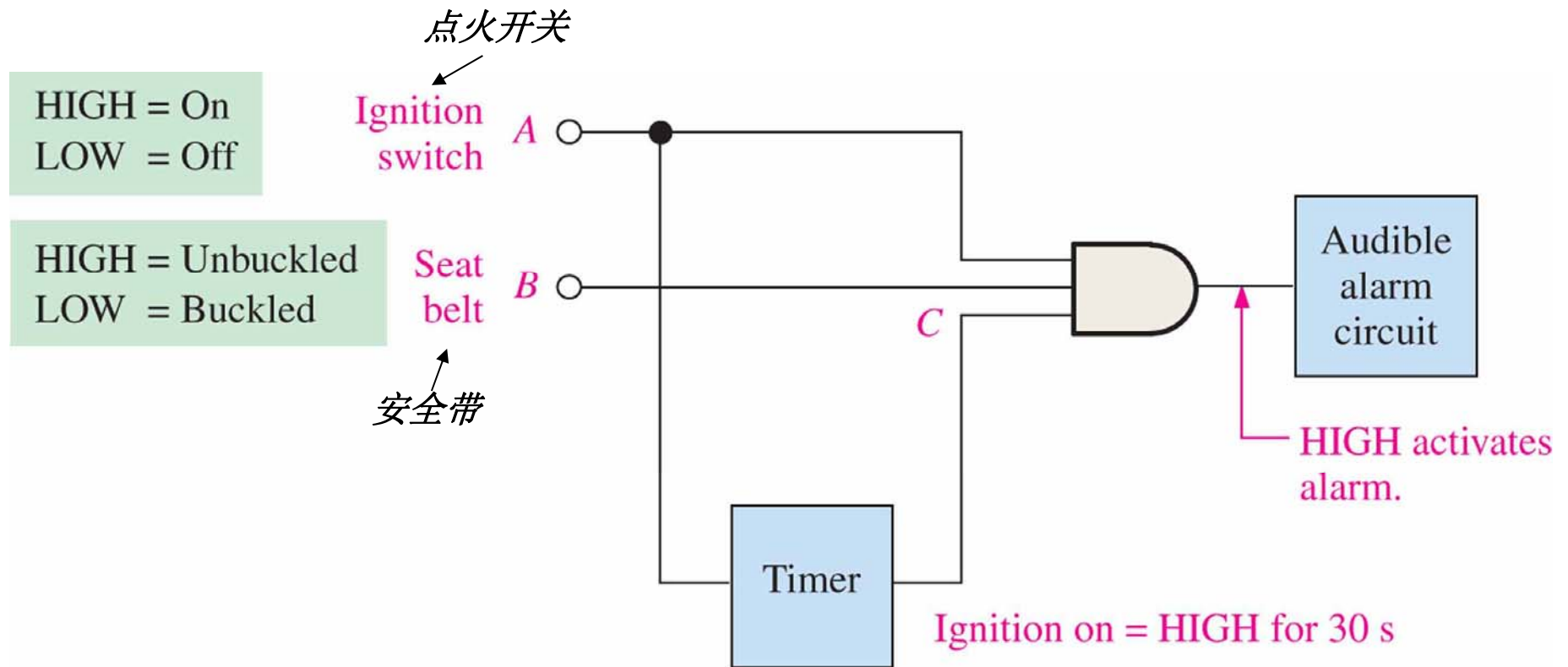
The AND Gate can be extended to any number of inputs



An AND gate performing an enable/inhibit function for a frequency counter.

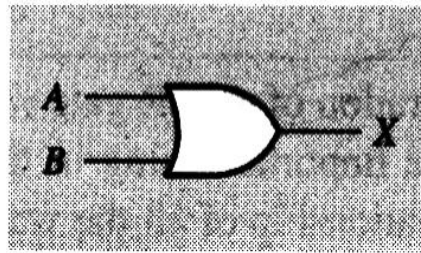


A simple seat belt alarm circuit using an AND gate.

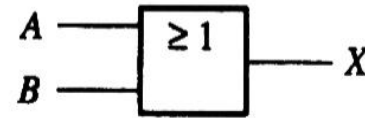


3-3 THE OR GATE (或门)

- The OR gate produces a **LOW** only when **all of the inputs are LOW**. When any of the inputs is **HIGH**, the output is **HIGH**.
- The basic purpose of an OR gate is to determine when one or more conditions are true.



(a) Distinctive shape



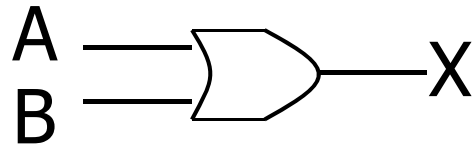
(b) Rectangular outline with the OR (≥ 1) qualifying symbol

FIGURE 3-17

Standard logic symbols for the OR gate showing two inputs (ANSI/IEEE Std. 91-1984).

The OR Gate

•**Symbol**



•**Logic expression**

$$X = A + B$$

•**Truth table**

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

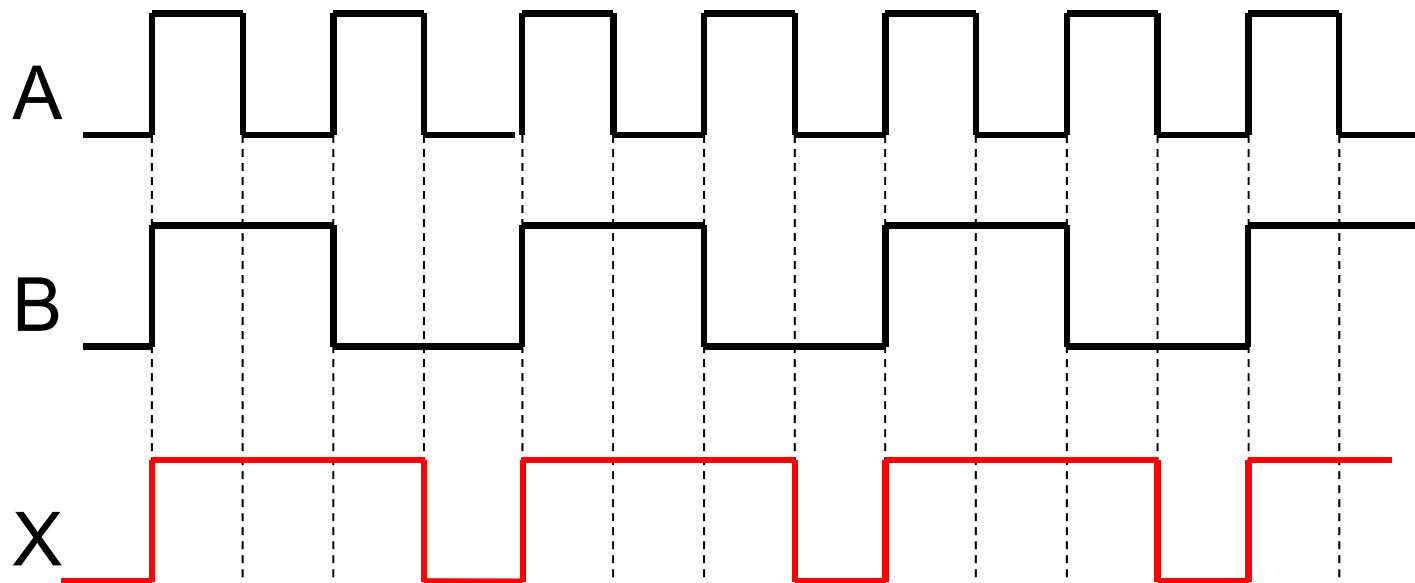
The OR Gate

- Boolean addition is the same as the OR function. (和1-binary addition 不同)

TABLE 3-6

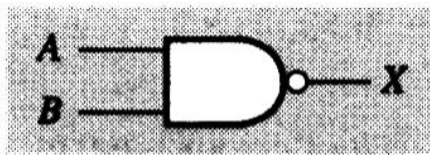
<i>A</i>	<i>B</i>	$A + B = X$
0	0	$0 + 0 = 0$
0	1	$0 + 1 = 1$
1	0	$1 + 0 = 1$
1	1	$1 + 1 = 1$

• *Waveform*

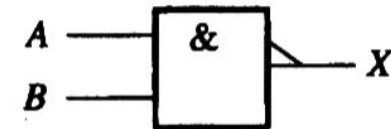


3-4 THE NAND GATE (与非门)

- The NAND gate produces a LOW only when all of the inputs are HIGH. When any of the inputs is LOW, the output is HIGH.
- The term NAND is a contraction of NOT-AND and implies an AND function with a complemented (inverted) output.



≡



(a) Distinctive shape, 2-input NAND gate and its NOT/AND equivalent

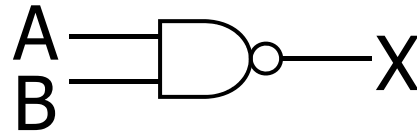
(b) Rectangular outline, 2-input NAND gate with polarity indicator

FIGURE 3-25

Standard NAND gate logic symbols (ANSI/IEEE Std. 91-1984).

The NAND Gate

• **Symbol**



Negative-OR

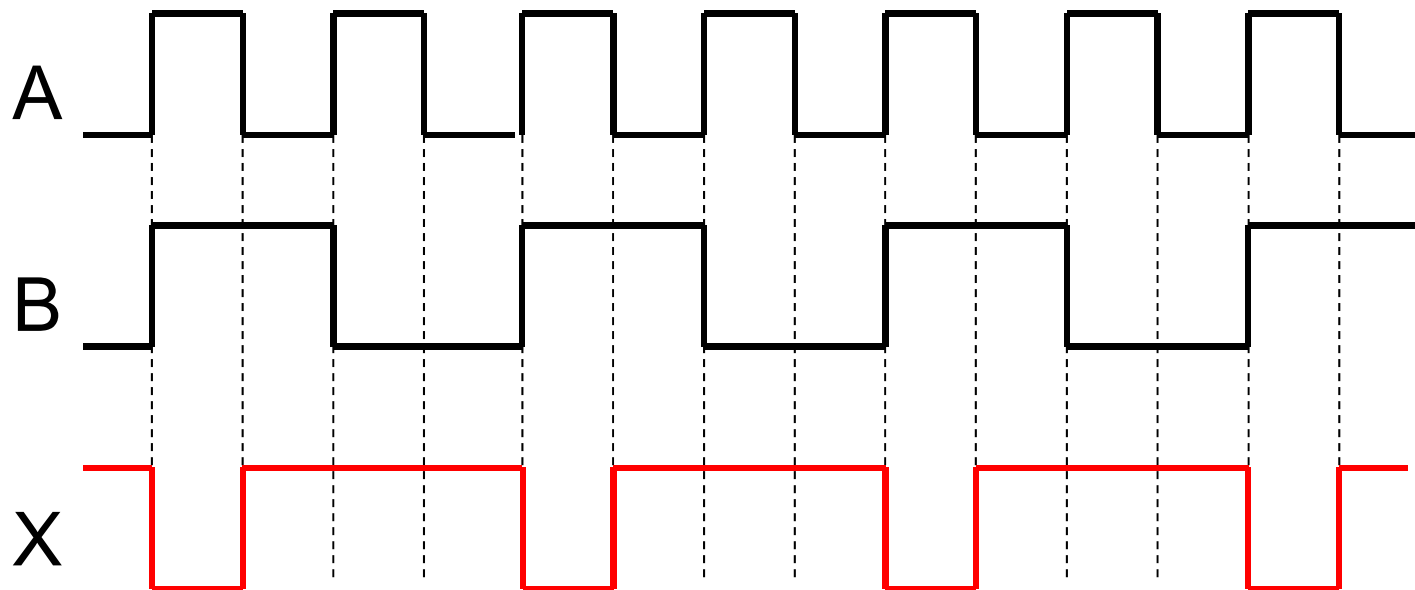
• **Logic expression**

$$X = \overline{AB}$$

• **Truth table**

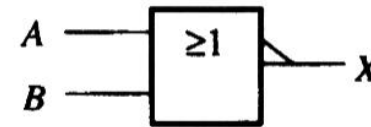
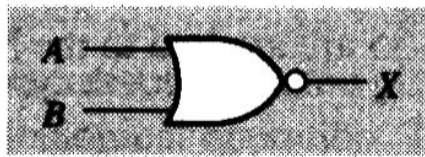
Inputs		Output
A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

• *Waveform*



3-5 THE NOR GATE (或非门)

- The NOR gate produces a HIGH only when all of the inputs are LOW. When any of the inputs is HIGH, the output is LOW.



(a) Distinctive shape, 2-input NOR gate and its NOT/OR equivalent

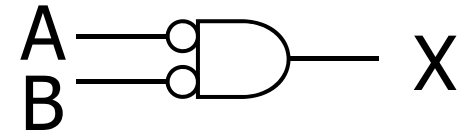
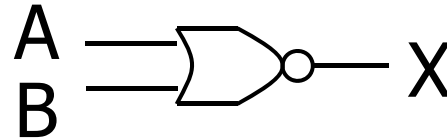
(b) Rectangular outline, 2-input NOR gate with polarity indicator

FIGURE 3-33

Standard NOR gate logic symbols (ANSI/IEEE Std. 91-1984).

The NOR Gate

- **Symbol**



- **Logic expression**

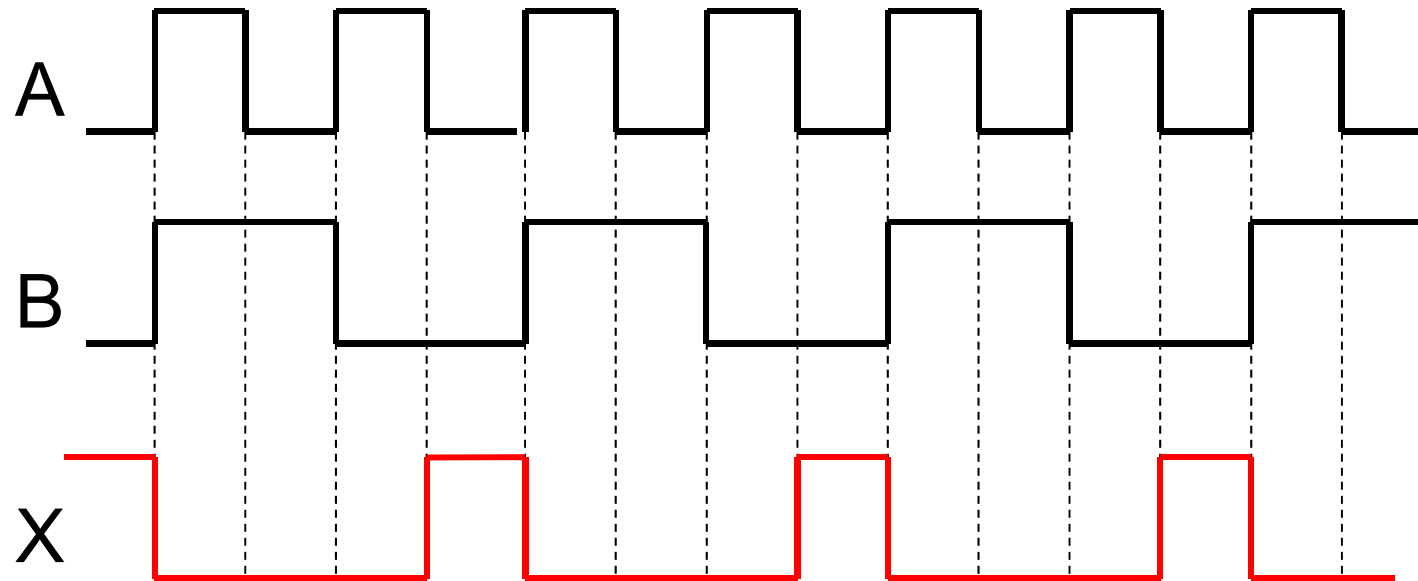
$$X = \overline{A + B}$$

Negative-AND gate

- **Truth table**

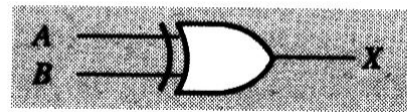
Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

- ***Waveform***

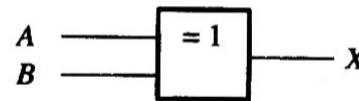


3-6 THE EXCLUSIVE-OR AND EXCLUSIVE-NOR GATES

- The exclusive-OR (XOR) (异或门) gate has only two inputs.
- The output of an XOR gate is HIGH only when two inputs are at opposite logic levels.



(a) Distinctive shape

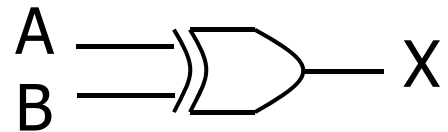


(b) Rectangular outline with the XOR qualifying symbol (= 1)

FIGURE 3-41
Standard logic symbols for the exclusive-OR gate.

The XOR Gates (异或门)

- **Symbol**



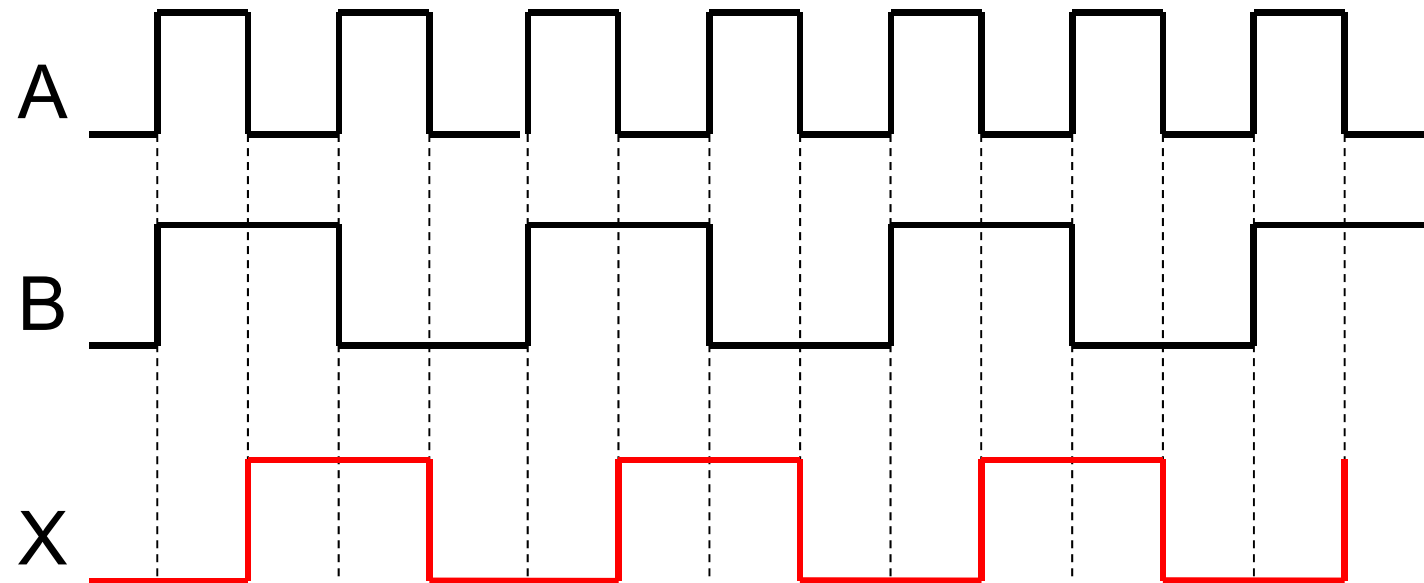
- **Logic expression**

$$X = A \oplus B = \overline{A}B + A\overline{B}$$

- **Truth table**

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

• *Waveform*

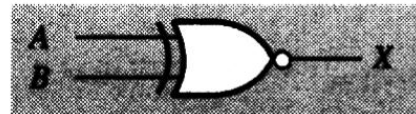


The XNOR gates (同或门)

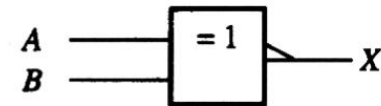
- The exclusive-NOR (XNOR) gate has only two inputs.
- The output of an XNOR gate is LOW only when two inputs are at opposite logic levels.

FIGURE 3-44

Standard logic symbols for the exclusive-NOR gate.



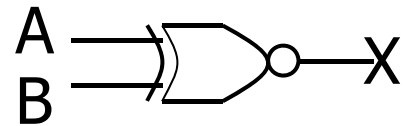
(a) Distinctive shape



(b) Rectangular outline

The XNOR Gates

- **Symbol**



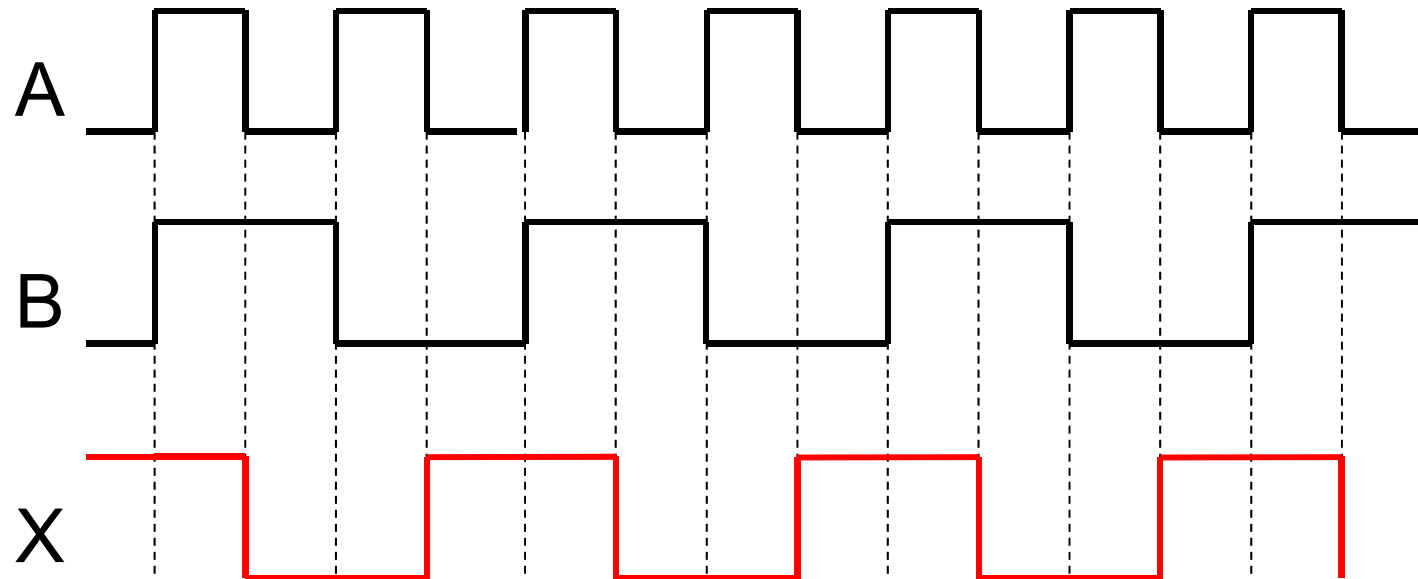
- **Logic expression**

$$X = \overline{A \oplus B} = AB + \overline{A}\overline{B} = A \odot B$$

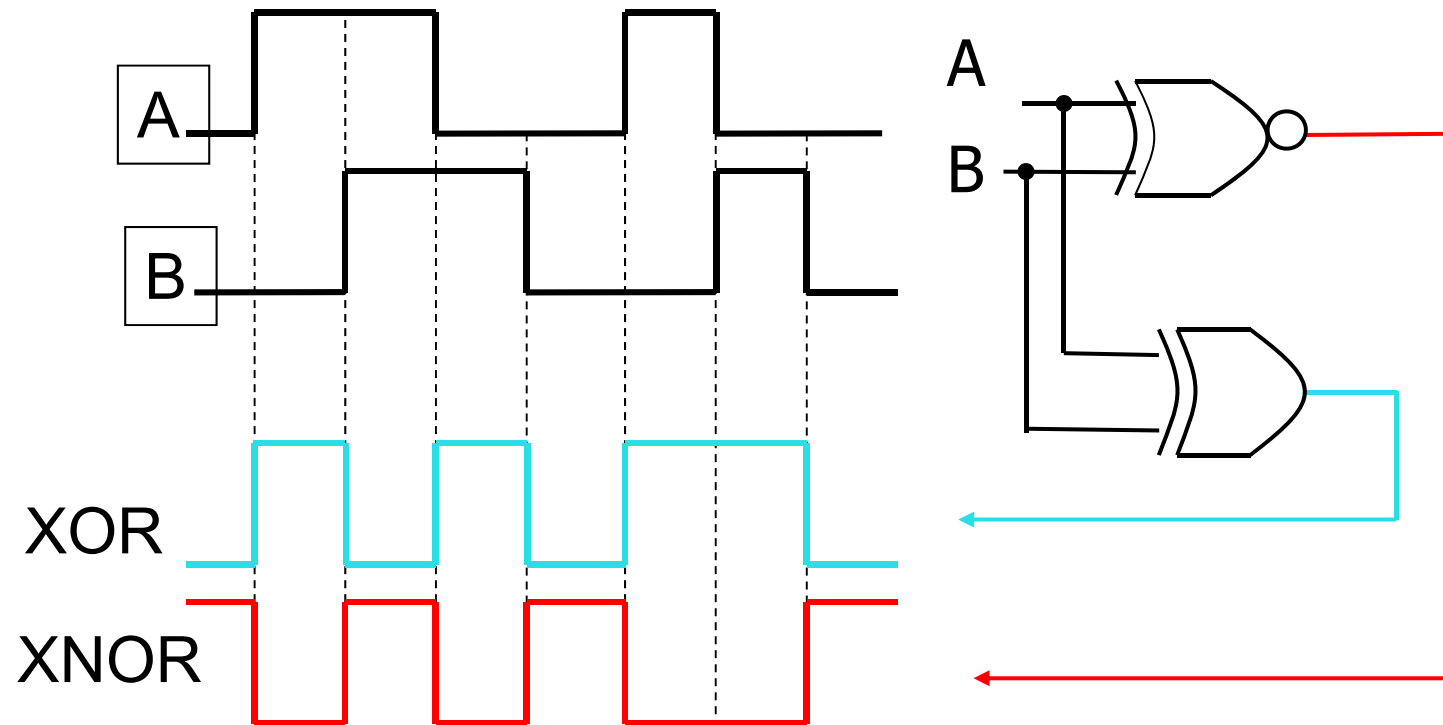
- **Truth table**

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

• *Waveform*

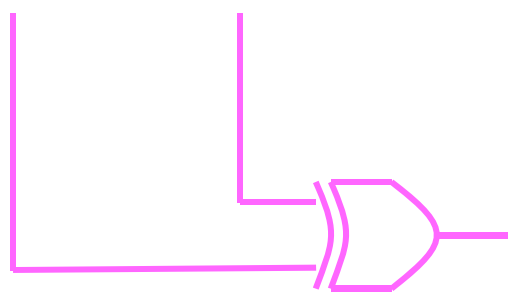


Example: Determine the output waveform for the XOR gate and for the XNOR, given the input waveforms of A and B as follows:



Example: The XOR gate used to add two bits.

Input bits		Output
A	B	Sum
0	0	0
0	1	1
1	0	1
1	1	0 (without 1 carry)



Quiz

1. The truth table for a 2-input AND gate is

a.

Inputs		Output
<i>A</i>	<i>B</i>	<i>X</i>
0	0	0
0	1	1
1	0	1
1	1	0

c.

Inputs		Output
<i>A</i>	<i>B</i>	<i>X</i>
0	0	0
0	1	0
1	0	0
1	1	1

b.

Inputs		Output
<i>A</i>	<i>B</i>	<i>X</i>
0	0	1
0	1	0
1	0	0
1	1	0

d.

Inputs		Output
<i>A</i>	<i>B</i>	<i>X</i>
0	0	0
0	1	1
1	0	1
1	1	1

Quiz

2. The truth table for a 2-input NOR gate is

a.

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

c.

Inputs		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

b.

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

d.

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

Quiz

3. The truth table for a 2-input XOR gate is

a.

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

b.

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

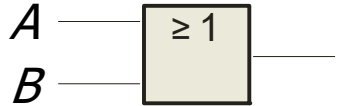
c.

Inputs		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

d.

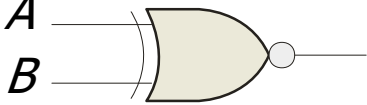
Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

Quiz

4. The symbol  is for a(n)

- a. OR gate
- b. AND gate
- c. NOR gate
- d. XOR gate

Quiz

5. The symbol  is for a(n)

- a. OR gate
- b. XNOR gate
- c. NOR gate
- d. XOR gate

Quiz

6. A logic gate that produces a HIGH output only when all of its inputs are HIGH is a(n)

- a. OR gate
- b. AND gate
- c. NOR gate
- d. NAND gate

Quiz

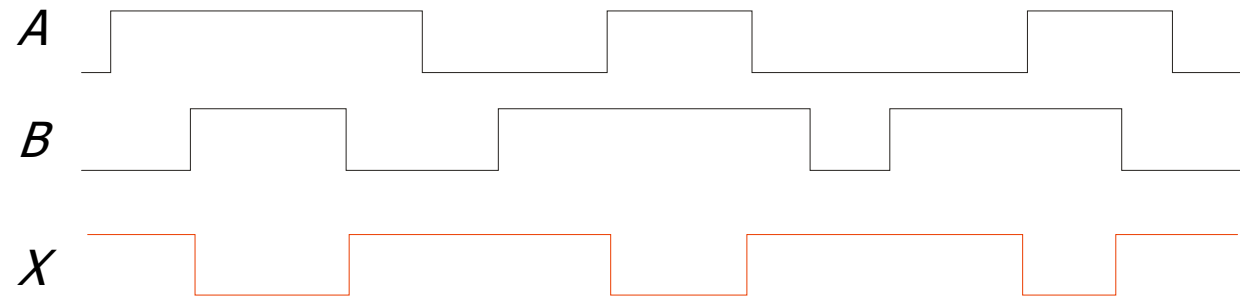
7. The expression $X = A + B$ means

- a. $A \text{ OR } B$
- b. $A \text{ AND } B$
- c. $A \text{ XOR } B$
- d. $A \text{ XNOR } B$

Quiz

8. A 2-input gate produces the output shown. (X represents the output.) This is a(n)

- a. OR gate
- b. AND gate
- c. NOR gate
- d. NAND gate



Quiz

9. A 2-input gate produces a HIGH output only when the inputs agree. This type of gate is a(n)

- a. OR gate
- b. AND gate
- c. NOR gate
- d. XNOR gate

Homework

- Problems: 2, 8, 13, 15, 20, 24, 28



Answers

1. c 6. b

2. b 7. a

3. a 8. d

4. a 9. d

5. b